FIRST INTERIM REPORT

Data-Driven and Physics-Based Thermal Modeling of Building Envelopes in Iceland

Brendan Rudberg | Caltech SURF, hosted at the University of Iceland

Supervisor: Prof. Ólafur Pétur Pálsson (University of Iceland). Co-supervised by Dipendra Gautam (PhD).

July 2026

1. Motivation and Background

State-space modeling of building thermal dynamics is a mature field with two dominant and largely separate modeling philosophies. Grey-box models pair a physically motivated structure, typically a lumped-parameter resistance-capacitance (RC) network, with statistical parameter estimation. Because their states and parameters correspond to physical quantities such as thermal resistances and heat capacities, they support qualitative analysis, physical reasoning, and the inference of causal structure rather than mere correlation. Black-box models, most prominently neural networks, impose little physical structure and instead learn input-output relationships directly from data. They are easy to deploy and can capture complex, unmodeled relationships, at the cost of interpretability.

The research under Prof. Pálsson has a long history in stochastic grey-box modeling of the heat dynamics of buildings, work rooted in the continuous-time stochastic differential equation framework developed with Henrik Madsen and collaborators at DTU [1, 2, 4]. The RC-network identification approach and the forward model-selection strategy of Bacher and Madsen [3] are standard tools in this line of research. In parallel, the wider building-energy community has increasingly adopted data-driven and neural methods [5], and hybrid, physics-informed approaches are now emerging [6].

Despite the maturity of each approach, there is limited work that systematically compares grey-box and black-box state-space methods on the same large, well-controlled dataset. Most studies commit to a single modeling family, which makes it difficult to weigh interpretability against predictive power on equal terms. This project addresses that gap by generating a long, well-characterized record for one physical plant and using it to benchmark the two approaches directly.

2. Problem Statement and Approach

The plant under study is approximately one-dimensional heat flow through an exterior concrete wall and the adjacent air on either side. This is a clean, low-order thermal system whose grey-box representation, a small RC network with a handful of states, is well understood. That known physical baseline is what makes it a good testbed for comparing modeling methods.

Four temperature sensors instrument the system: outside air, outside wall surface, inside wall surface, and inside air. These four measurements map directly onto the node temperatures of a candidate RC model, so the same dataset can be used both to fit a physically interpretable grey-box model and to train black-box models, and the two can then be compared on predictive accuracy, data requirements, and interpretability.

The project has two phases. The first, and the subject of this report, is the design and validation of the data-acquisition system together with the collection of a continuous, multi-week temperature record. The second is the modeling comparison itself: fitting grey-box RC and state-space models alongside black-box neural models, then evaluating them on prediction accuracy, physical consistency, and robustness. Establishing a reliable and well-understood data pipeline in phase one is a prerequisite for a fair comparison in phase two, which is why the first weeks focus on acquisition.

3. Methods: Data Acquisition

Data are collected with a Raspberry Pi 4B reading four DS18B20 digital temperature sensors over the 1-Wire bus. All four sensors sit near an exterior wall of an isolated second-floor room to keep the measured system as close to one-dimensional as possible. The two air sensors hang roughly 12 inches from the wall on the inside and outside faces. The two wall-surface sensors are fixed to the inner and outer wall faces with putty and are fully insulated from the surrounding air with additional putty, so that each reads wall-surface temperature rather than a blend of wall and air. The wall is concrete and approximately 12 inches thick.

A logging script writes one row per minute containing the four temperatures and a Unix timestamp (time.time) to a CSV file. A MATLAB script then reads the CSV, converts the timestamps back to calendar date and time, and plots the four channels for inspection.

4. Progress to Date

The data-acquisition system is built and operating. The Raspberry Pi, the four DS18B20 sensors, and the 1-Wire wiring are assembled and mounted at the wall, and the per-minute CSV logging and the MATLAB plotting pipeline are both working end to end. Roughly one week of continuous, minute-resolution data across all four channels has been collected.

The recorded channels behave as physically expected: the two air temperatures track the room and outdoor conditions, the two wall-surface temperatures are smoother and lag the air, and the ordering of the four signals across the wall is consistent with the expected temperature gradient. The first day of data looks anomalous, most likely reflecting initial thermal settling of the putty-mounted sensors and early adjustments to the setup, while the remainder of the week looks clean and usable. Figure 1 shows the full four-channel record for the collection week.

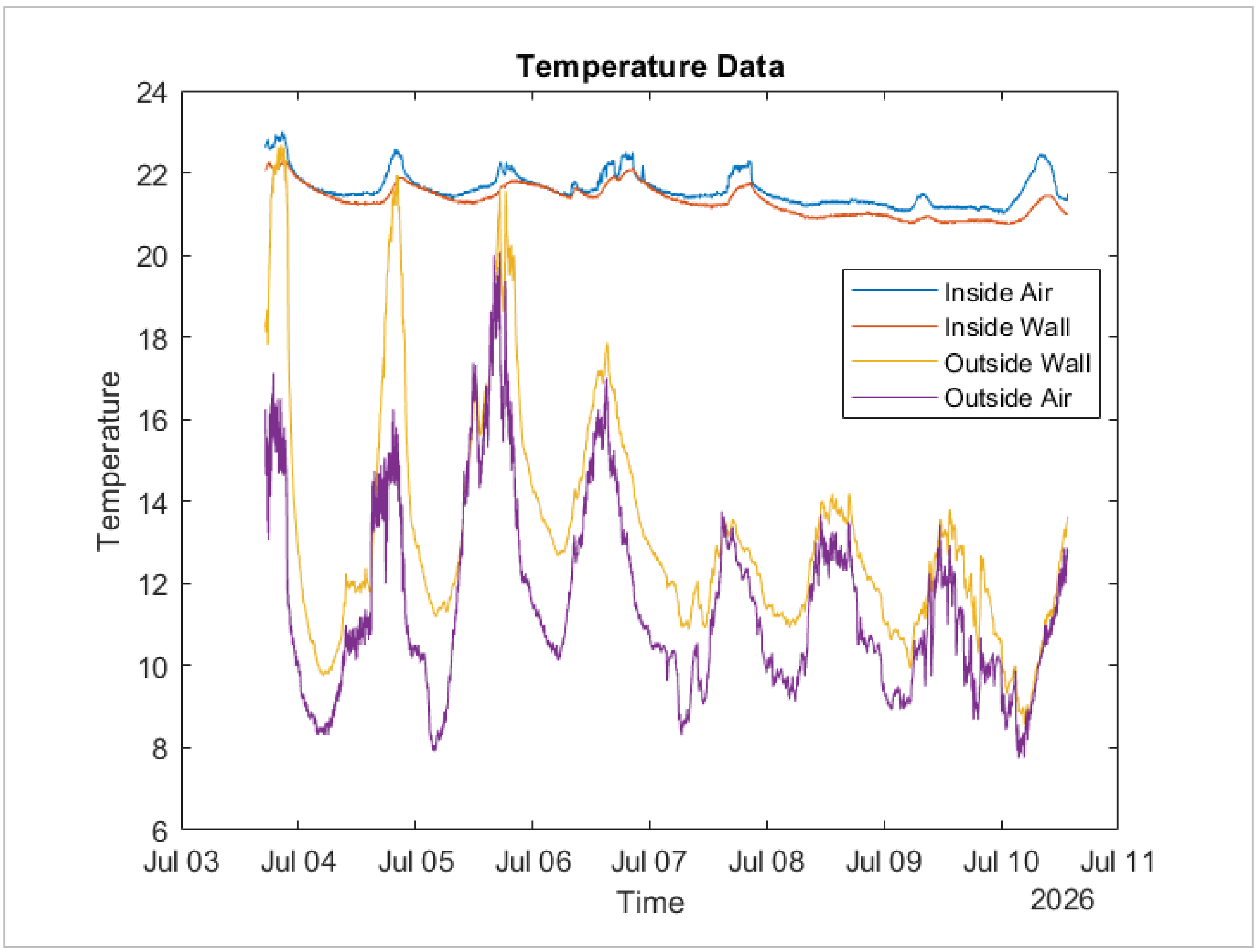

Figure 1. Outside air, outside wall, inside wall, and inside air temperatures over the collection week.

5. Challenges and Anticipated Difficulties

The most disruptive challenge has been the lack of a working internet connection at the Raspberry Pi, caused by ethernet and wifi problems in the building. This prevents remote access, so data currently has to be retrieved manually, and it also blocks the automatic collection of external covariates such as humidity, solar radiation, and wind. Limited access to materials on site made the sensor wiring more difficult than expected. A further complication is confounding disturbances from the building itself: radiators cycling on and off in other rooms inject heat that the four sensors do not directly observe.

Moving forward, the absence of weather covariates is the main anticipated limitation. Without solar, wind, and humidity inputs, the achievable accuracy of both model families may be capped, and a local logging solution or a nearby reference weather station may be needed. The radiator-driven disturbances will complicate causal attribution when fitting the grey-box model, since unmeasured heat inputs can be absorbed into the estimated parameters. It will also be important to collect enough data, with sufficient thermal excitation, for the black-box models to be trained and compared against the grey-box models on fair terms rather than being starved of the variation they need to learn.

References

[1] Madsen, H., and Holst, J. (1995). Estimation of continuous-time models for the heat dynamics of a building. Energy and Buildings, 22(1), 67-79.

[2] Andersen, K. K., Madsen, H., and Hansen, L. H. (2000). Modelling the heat dynamics of a building using stochastic differential equations. Energy and Buildings, 31(1), 13-24.

[3] Bacher, P., and Madsen, H. (2011). Identifying suitable models for the heat dynamics of buildings. Energy and Buildings, 43(7), 1511-1522.

[4] Kristensen, N. R., Madsen, H., and Jørgensen, S. B. (2004). Parameter estimation in stochastic grey-box models. Automatica, 40(2), 225-237.

[5] Amasyali, K., and El-Gohary, N. M. (2018). A review of data-driven building energy consumption prediction studies. Renewable and Sustainable Energy Reviews, 81, 1192-1205.

[6] Drgoňa, J., Tuor, A. R., Chandan, V., and Vrabie, D. L. (2021). Physics-constrained deep learning of multi-zone building thermal dynamics. Energy and Buildings, 243, 110992.